

Nmap TCP SYN Scan Report

1. Introduction

Network scanning is one of the first steps in cybersecurity and penetration testing. It helps identify active devices connected to a network, discover open ports, and determine which services are running. Understanding these services allows security professionals to identify potential vulnerabilities and improve network security.

2. Objective

The objectives of this assignment were:

- Install Nmap from the official website.
- Identify the local IP address and network range.
- Perform a TCP SYN scan.
- Discover active hosts.
- Identify open TCP ports.
- Research services running on those ports.
- Identify potential security risks.
- Save scan results for documentation.

3. Tools Used

- Nmap 7.99
- Windows Command Prompt
- Windows 11
- Mobile Hotspot Wi-Fi

4. Network Information

IPv4 Address

10.77.196.176

Subnet Mask

255.255.255.0

Target Network

10.77.196.0/24

5. Scan Command

```
nmap -sS 10.77.196.0/24
```

To save the scan result:

```
nmap -sS 10.77.196.0/24 -oN hotspot_scan.txt
```

6. Scan Results

IP Address	Status	Open Ports	Service
10.77.196.46	Host Up	None	All scanned ports closed
10.77.196.70	Host Up	2869	ICSLAP
10.77.196.184	Host Up	None	All scanned ports closed
10.77.196.198	Host Up	53	DNS
10.77.196.176	Host Up	135,139,445,5432	MSRPC, NetBIOS, SMB, PostgreSQL

7. Service Analysis

Port 53 – DNS

Purpose:

Provides domain name resolution between hostnames and IP addresses.

Security Risk:

A misconfigured DNS service can be abused for unauthorized queries or amplification attacks.

Recommendation:

Restrict access to trusted devices and keep DNS services updated.

Port 135 – Microsoft RPC

Purpose:

Allows communication between Windows applications and services.

Security Risk:

Can become a target if Windows systems are not patched.

Recommendation:

Keep Windows updated and block unnecessary external access.

Port 139 – NetBIOS Session Service

Purpose:

Supports Windows file and printer sharing.

Security Risk:

May expose shared resources and system information.

Recommendation:

Disable NetBIOS if it is not required.

Port 445 – Microsoft-DS (SMB)

Purpose:

Provides Windows file and printer sharing.

Security Risk:

SMB has been exploited by malware such as WannaCry.

Recommendation:

Disable SMBv1, keep Windows patched, and restrict SMB traffic using a firewall.

Port 5432 – PostgreSQL

Purpose:

Default port used by PostgreSQL database servers.

Security Risk:

Weak passwords or unnecessary exposure may lead to unauthorized database access.

Recommendation:

Use strong authentication, restrict network access, and keep PostgreSQL updated.

Port 2869 – ICSLAP

Purpose:

Associated with UPnP and Windows device discovery services.

Security Risk:

If unnecessary, the service increases the attack surface of the host.

Recommendation:

Disable UPnP or related services when they are not required.

8. Security Risks Summary

- Windows networking services should only be exposed when necessary.
- Database ports should never be publicly accessible without authentication.
- DNS services should be restricted to trusted clients.
- Unused services should be disabled to reduce the attack surface.
- Firewalls and regular software updates significantly improve security.

9. Conclusion

The TCP SYN scan successfully identified active hosts and detected several open TCP ports within the local network. The services associated with these ports were analyzed, and potential security risks were identified. This exercise provided practical experience with network reconnaissance, service identification, and basic security assessment using Nmap.

10. Screenshots Included

- Nmap installation verification (nmap --version)
- IP configuration (ipconfig)
- Nmap TCP SYN scan output